

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Piping Wall Thickness Calculation
Document Number	EA-BFPCL-GPMC-GGPL-FEED-MPP-RPT-005
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Piping Wall Thickness Calculation

R01	01.06.22	Issued for Review	E. Omeke	E. Nwachukwu	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	01-06-22	Issued for Client's Review
R02		
A01		

Review Code & Status (to be filled by Client)

☐ Code 1	Approved – Further work may proceed.	
☐ Code 2	Approved with comments. – Incorporate Comments. Up-Rev & Resubmit. Work May Proceed	
☐ Code 3	Reviewed – Incorporate Comments. Up-Rev & Resubmit. Work May Not Proceed	
☐ Code 4	Retained for information only	
☐ Code 5	Review Not Required – Do not Resubmit	
Date:	Position:	Sign:

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-MPP-RPT-005

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-015	Specification for Piping
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005	Piping Classes
EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003 Sheet 1 to 8	Process PIDs

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Abbreviations

Abbreviations	Meaning
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
API	American Petroleum Institute
BS	British Standard
CS	Carbon Steel
ENP	Electroless Nickel Plated
ISO	International Organisation for Standardization
LTCS	Low Temperature Carbon Steel
MESC	Material & Equipment Standard and Code
Mmscf/d	Million standard cubic feet per day
MSS	Manufacturers Standardisation Society
MTO	Material Take Off
NDE	Non-Destructive Examination
NPS	Nominal Pipe Size
SPE	Society of Petroleum Engineers
SS	Stainless Steel

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

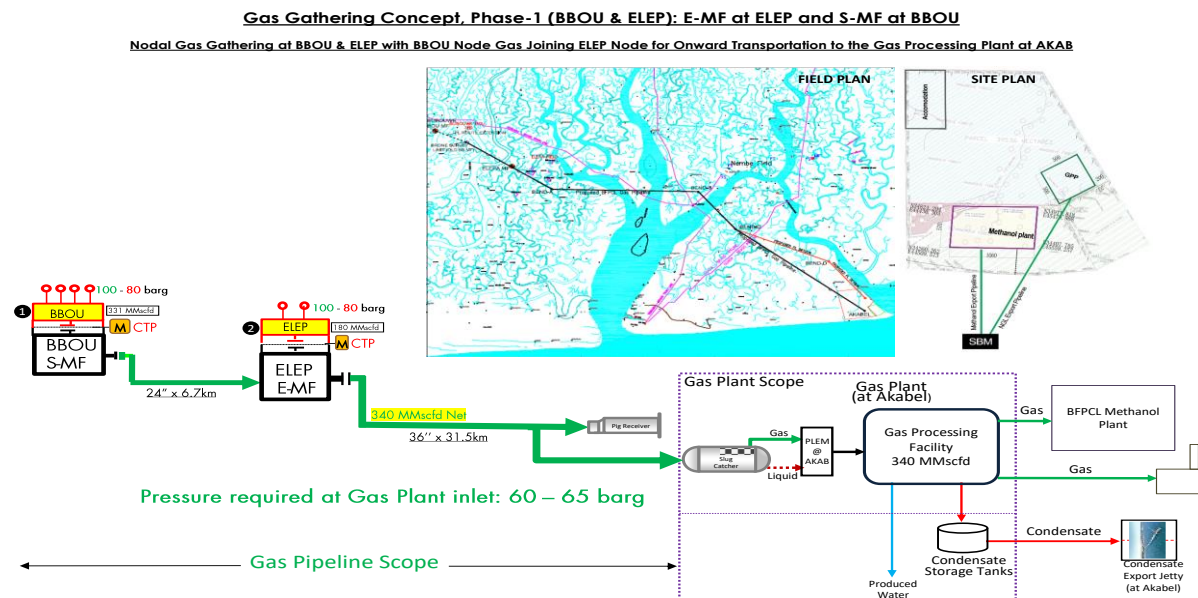


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This document presents calculations performed to determine the wall thickness for various process lines for all associated piping at BBOU Manifold, ELP Manifold and AKABEL end facilities of BFPCL Gas Gathering Pipelines Project (Phase 1).

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 31.38.01.25-Gen.	Piping – Process Design Requirements
Dep 31.38.01.10-Gen	Piping Classes – Basis of Design
DEP 31.38.01.24-Gen.	Piping – Engineering and Layout Requirements
DEP 31.38.01.31-Gen	Shop And Field Fabrication of Piping
DEP 31.38.01.84-Gen	Piping Classes
ASME B1.1	Unified Inch Screw Threads (UN and UNR Thread Form)
ASME B1.20.1	Pipe Threads, General Purpose
ASME B16.5	Pipe Flanges and Flanged Fittings
ASME B16.9	Factory Made Wrought Steel Butt Welding Fittings
ASME B16.11	Forged Steel Fittings, Socket Welding and Threaded
ASME B16.25	Butt Welding Ends
ASME B31.8	Gas transmission and Distribution Piping Systems
ASME B31.3	Process Piping
ASME B36.10M	Welded and Seamless Wrought Steel Pipe
ASTM A106	Seamless Carbon Steel Pipe for High Temperature Service
ASTM A333	Standard Specification for Seamless and Welded Steel Pipe for Low-Temperature Service and Other Applications with Required Notch Toughness
API 5L	Specification for Line Pipe

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Methodology

The design code is ASME B31.3 and the delineation of the process piping is as defined in the PIDs referenced in this document.

In place of using the project design conditions, Maximum values of wall thicknesses at temperature and pressure combinations defined in ASME B16.5 were calculated for. This is to ensure that the wall thickness selected shall be capable of withstanding the loads exerted on the wall of the pipeline by the process fluid internal pressure over its service life and over several range of design and operating conditions.

For the project, wall thickness for NPS 2 inches and below shall not be less than SCH 80.

Commercial pipe schedule selection is as per ASME B36.10/B36.10M.

Carbon steel Material selected for moderate temperature in plot piping is ASTM A106 GR. B or API 5L Gr B for NP 1/2" to 16" and API 5L Gr. X60 for NP 18" to 46"

Carbon steel Material selected for low temperature in plot piping is ASTM A333 Gr. 6

The Material for Stainless Steel is ASTM A312 TP 316L and DUPLEX S31803

3.1. Piping Wall thickness Calculation under ASME B31.3 Code

- The general guidelines for pressure design detailed in section 304 of the ASME B31.3 requirements formed the basis for calculating minimum pipe design wall thickness
- Erosion allowance is not considered in the wall thickness calculation
- A corrosion allowance of 3mm applicable to BBOU station and 6mm applicable to ELEPA and AKABEL were applied in the calculations.

For $t < D/6$, or $P/SE > 0.385$ minimum design wall thickness for straight pipe under internal pressure is calculated using the following equations (ASME B31.3, Section 304.1.2):

$$t_m = t + c \quad , \text{ and} \quad \text{Equation 1}$$

$$t = \frac{PD}{2(SEW + PY)} \quad \text{Equation 2}$$

Where,

t_m =Minimum required thickness, including mechanical, corrosion allowances, mm

t =Pressure design wall thickness, mm

P=Design pressure based on pressure ceiling value for specific material grade as defined in ASME B16.5, kPa

D=Outside diameter of pipe per applicable dimension standard.

S=Stress value for material from Table A-1 of ASME B31.3, kPa

E=Quality factor from Table A-1A or A-1B of ASME B31.3

W=Weld joint strength reduction factor per parameter, obtained from Section 302.3.5 (e) of ASME B31.3

Y=Y is the coefficient based on ASME B31.3, table 304.1.1, valid for $t < D/6$ and for materials shown. The value of Y may be interpolated for intermediate temperatures. $t \geq D/6$, the coefficient should be calculated using:

$$Y = \frac{d + 2c}{D + d + 2c} \quad \text{Equation 3}$$

- d = Inside diameter of the pipe. For pressure design calculations, the inside diameter of the pipe is the maximum value allowable under purchase specifications.
- c = Sum of the mechanical allowances (threaded or groove depth) plus corrosion allowances.
- m = Coefficient to account for negative manufacturing tolerance when pipe is ordered by its nominal wall thickness, calculated as follows:

$$m = \frac{100}{100 - a} \quad \text{Equation 4}$$

- a = Percentage negative manufacturing tolerance, where a is
 - Minus 12 ½ %

T = Pipe wall thickness, calculated as follows:

$$T = (t_m)(m) = t_m \frac{100}{100 - a} \quad \text{Equation 5}$$

4. Results

The results are presented in appendix 1 below comprising of tables for the following Piping Classes.

Piping class	Pressure Class	Material Type	Material Grade	Corrosion Allowance
B03	150#	Carbon Steel	API 5L Gr. B	3mm
B06	150#	Carbon Steel	API 5L Gr. B	3mm
B13	300#	Carbon Steel	API 5L Gr. B	3mm
B23	150#	Low Temperature Carbon Steel	A333 Gr. 6	3mm
C03	900#	Carbon Steel	API 5L Gr. B	3mm
C03	900#	Carbon Steel	API 5L Gr. X60	3mm
C06	900#	Carbon Steel	API 5L Gr. B	6mm
C06	900#	Carbon Steel	API 5L Gr. X60	6mm
C23	900#	Low Temperature Carbon Steel	A333 Gr. 6	6mm
C50	900#	Stainless Steel	A312 TP316L	0mm
C70	900#	DUPLEX Stainless Steel	A790 S31803	0mm
D70	1500#	DUPLEX Stainless Steel	A790 S31803	0mm

Appendix 1: Piping Wall Thickness Calculation Tables

WALL THICKNESS CALCULATION FOR MATERIAL API 5LGr B (B03-150#) PER ASME B 31.3-2018

	No.		DESIGN DATA SUMMARY	NOTATION		INPUT	UNIT	REMARK
	1	Design	Design Temperature		>>	38	°C	ATM Temp = 38°C
	2	Condition	Design Pressure	P	>>	19	Bar	
	3	Pipe Data	Pipe Material		>>	API 5L Gr. B		
	4		Pipe Allowable Stress at Design Temp. 38°C	S _{oper}	=	206.84	MPa	
			Pipe Allowable Stress at ATM Temp. 38°C	S _{atm}	=	137.90	MPa	
	5	Allowance	Corrosion Allowance, Bore	C.A	>>	3.0	mm	
	6		Thread Allowance, Bore	M.A	>>	0.00	mm	
	7	Factor	Longitudinal Weld Joints Factor	E	>>	1.00		
	8		Temperature Coefficient	Y	>>	0.4		
	9		Weld Joint Strength Reduction Factor	W	>>	1.00		
	10	Tolerance	Mill Tolerance		>>	12.5	%	

PROJECT TITLE : Front End Engineering Design for Gas Gathering Pipelines (Phase1)
PROJECT No : 11998A
PIPE CLASS : B03

	Pipe Size		Outside Diameter	Design Pressure	Material	Temp	Allowable Stress	Yield Stress	E Factor	W (Weld Joint Strength Reduction Factor)	Y Coefficient	Corrosion Allowance	Mech. Allowance (Thread/Groove Depth)	Mill. Tol	Pressure Design Thickness	Required Thickness	Minimum Thickness	Selected Schedule / Thickness		Selected Wall Thickness Validation	Para 304.1.2 (a) Equation (3a) Validation	
			D	P			S _o	S _y	E	W	Y	C.A	M.A	Tol.	t	t _m	T	T _s		T _s ≥ T	t < D/6	P _{/SE} ≤ 0.385
	NPS	DN	mm	kpa		°C	kpa	kpa				mm	mm	%	mm	mm	mm	Schedule >>	mm	OK / NOT OK	OK / NOT OK	OK / NOT OK
	½	15	21.3	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	0.10	3.10	3.54	160	4.78	OK	OK	OK
	¾	20	26.7	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	0.12	3.12	3.57	160	5.56	OK	OK	OK
	1	25	33.4	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	0.15	3.15	3.60	160	6.35	OK	OK	OK
	1½	40	48.3	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	0.22	3.22	3.68	160	7.14	OK	OK	OK
	2	50	60.3	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	0.28	3.28	3.74	XS	5.54	OK	OK	OK
	3	80	88.9	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	0.41	3.41	3.89	STD	5.49	OK	OK	OK
	4	100	114.3	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	0.52	3.52	4.03	STD	6.02	OK	OK	OK
	6	150	168.3	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	0.77	3.77	4.31	STD	7.11	OK	OK	OK
	8	200	219.1	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	1.00	4.00	4.57	20	6.35	OK	OK	OK
	10	250	273.1	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	1.25	4.25	4.86	20	6.35	OK	OK	OK
	12	300	323.9	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	1.48	4.48	5.12	30	8.38	OK	OK	OK
	14	350	355.6	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	1.63	4.63	5.29	20	7.92	OK	OK	OK
	16	400	406.4	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	1.86	4.86	5.55	20	7.92	OK	OK	OK
	18	450	457.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	2.09	5.09	5.82	20	7.92	OK	OK	OK
	20	500	508.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	2.32	5.32	6.09	STD	9.53	OK	OK	OK
	24	600	610.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	2.79	5.79	6.62	STD	9.53	OK	OK	OK
	26	650	660.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	3.02	6.02	6.88	STD	9.53	OK	OK	OK
	28	700	711.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	3.25	6.25	7.15	STD	9.53	OK	OK	OK
	30	750	762.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	3.49	6.49	7.41	STD	9.53	OK	OK	OK
	32	800	813.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	3.72	6.72	7.68	40	10.31	OK	OK	OK
	34	850	864.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	3.95	6.95	7.95	XS	12.7	OK	OK	OK
	36	900	914.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	4.18	7.18	8.21	XS	12.7	OK	OK	OK
	38	950	965.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	4.42	7.42	8.48	XS	12.7	OK	OK	OK
	40	1000	1016.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	4.65	7.65	8.74	XS	12.7	OK	OK	OK
	42	1050	1067.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	4.88	7.88	9.01	XS	12.7	OK	OK	OK
	44	1100	1118.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	5.12	8.12	9.28	XS	12.7	OK	OK	OK
	46	1150	1168.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	5.34	8.34	9.54	XS	12.7	OK	OK	OK

WALL THICKNESS CALCULATION FOR MATERIAL API 5LGr B (B06-150#) PER ASME B 31.3-2018

	No.		DESIGN DATA SUMMARY	NOTATION		INPUT	UNIT	REMARK
	1	Design	Design Temperature		>>	38	°C	ATM Temp = 38°C
	2	Condition	Design Pressure	P	>>	19	Bar	
	3	Pipe Data	Pipe Material		>>	API 5L Gr. B		
	4		Pipe Allowable Stress at Design Temp. 38°C	S _{oper}	=	206.84	MPa	
			Pipe Allowable Stress at ATM Temp. 38°C	S _{atm}	=	137.90	MPa	
	5	Allowance	Corrosion Allowance, Bore	C.A	>>	6.0	mm	
	6		Thread Allowance, Bore	M.A	>>	0.00	mm	
	7	Factor	Longitudinal Weld Joints Factor	E	>>	1.00		
	8		Temperature Coefficient	Y	>>	0.4		
	9		Weld Joint Strength Reduction Factor	W	>>	1.00		
	10	Tolerance	Mill Tolerance		>>	12.5	%	

PROJECT TITLE : Front End Engineering Design for Gas Gathering Pipelines (Phase1)																						
PROJECT No : 11998A																						
PIPE CLASS : B06																						
	Pipe Size		Outside Diameter	Design Pressure	Material	Temp	Allowable Stress	Yield Stress	E Factor	W (Weld Joint Strength Reduction Factor)	Y Coefficient	Corrosion Allowance	Mech. Allowance (Thread/Groove Depth)	Mill. Tol	Pressure Design Thickness	Required Thickness	Minimum Thickness	Selected Schedule / Thickness		Selected Wall Thickness Validation	Para 304.1.2 (a) Equation (3a) Validation	
			D	P			S _o	S _y	E	W	Y	C.A	M.A	Tol.	t	t _m	T	T _s		T _s ≥ T	t < D/6	^P / _{SE} ≤ 0.385
	NPS	DN	mm	kpa		°C	kpa	kpa				mm	mm	%	mm	mm	mm	Schedule >>	mm	OK / NOT OK	OK / NOT OK	OK / NOT OK
	½	15	21.3	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	0.10	6.10	6.97	XXS	7.47	OK	OK	OK
	¾	20	26.7	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	0.12	6.12	7.00	XXS	7.82	OK	OK	OK
	1	25	33.4	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	0.15	6.15	7.03	XXS	9.09	OK	OK	OK
	1½	40	48.3	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	0.22	6.22	7.11	XXS	10.15	OK	OK	OK
	2	50	60.3	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	0.28	6.28	7.17	XXS	11.07	OK	OK	OK
	3	80	88.9	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	0.41	6.41	7.32	160	11.13	OK	OK	OK
	4	100	114.3	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	0.52	6.52	7.45	120	11.13	OK	OK	OK
	6	150	168.3	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	0.77	6.77	7.74	XS	10.97	OK	OK	OK
	8	200	219.1	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	1.00	7.00	8.00	60	10.31	OK	OK	OK
	10	250	273.1	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	1.25	7.25	8.29	XS	12.7	OK	OK	OK
	12	300	323.9	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	1.48	7.48	8.55	40	10.31	OK	OK	OK
	14	350	355.6	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	1.63	7.63	8.72	40	11.13	OK	OK	OK
	16	400	406.4	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	1.86	7.86	8.98	40	12.7	OK	OK	OK
	18	450	457.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	2.09	8.09	9.25	30	11.13	OK	OK	OK
	20	500	508.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	2.32	8.32	9.51	XS	12.7	OK	OK	OK
	24	600	610.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	2.79	8.79	10.05	XS	12.7	OK	OK	OK
	26	650	660.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	3.02	9.02	10.31	XS	12.7	OK	OK	OK
	28	700	711.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	3.25	9.25	10.58	XS	12.7	OK	OK	OK
	30	750	762.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	3.49	9.49	10.84	60	14.27	OK	OK	OK
	32	800	813.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	3.72	9.72	11.11	30	15.88	OK	OK	OK
	34	850	864.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	3.95	9.95	11.38	30	15.88	OK	OK	OK
	36	900	914.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	4.18	10.18	11.64	30	15.88	OK	OK	OK
	38	950	965.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	4.42	10.42	11.90	-	15.88	OK	OK	OK
	40	1000	1016.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	4.65	10.65	12.17	-	15.88	OK	OK	OK
	42	1050	1067.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	4.88	10.88	12.44	-	15.88	OK	OK	OK
	44	1100	1118.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	5.12	11.12	12.70	-	15.88	OK	OK	OK
	46	1150	1168.0	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	5.34	11.34	12.97	80	17.48	OK	OK	OK

WALL THICKNESS CALCULATION FOR MATERIAL API 5LGr B (B13-150#) PER ASME B 31.3-2018

	No.		DESIGN DATA SUMMARY	NOTATION		INPUT	UNIT	REMARK
	1	Design	Design Temperature		>>	38	°C	ATM Temp = 38°C
	2	Condition	Design Pressure	P	>>	19	Bar	
	3	Pipe Data	Pipe Material		>>	API 5L Gr. B		
	4		Pipe Allowable Stress at Design Temp. 38°C	S _{oper}	=	206.84	MPa	
			Pipe Allowable Stress at ATM Temp. 38°C	S _{atm}	=	137.90	MPa	
	5	Allowance	Corrosion Allowance, Bore	C.A	>>	6.0	mm	
	6		Thread Allowance, Bore	M.A	>>	0.00	mm	
	7	Factor	Longitudinal Weld Joints Factor	E	>>	1.00		
	8		Temperature Coefficient	Y	>>	0.4		
	9		Weld Joint Strength Reduction Factor	W	>>	1.00		
	10	Tolerance	Mill Tolerance		>>	12.5	%	

PROJECT TITLE : [Front End Engineering Design for Gas Gathering Pipelines \(Phase1\)](#)
PROJECT No : [11998A](#)
PIPE CLASS : [B13](#)

Pipe Size			Outside Diameter	Design Pressure	Material	Temp	Allowable Stress	Yield Stress	E Factor	W (Weld Joint Strength Reduction Factor)	Y Coefficient	Corrosion Allowance	Mech. Allowance (Thread/Groove Depth)	Mill. Tol	Pressure Design Thickness	Required Thickness	Minimum Thickness	Selected Schedule / Thickness		Selected Wall Thickness Validation	Para 304.1.2 (a) Equation (3a) Validation	
			D	P			S _o	S _y	E	W	Y	C.A	M.A	Tol.	t	t _m	T	T _s		T _s ≥ T	t < D/6	P _{/SE} ≤ 0.385
	NPS	DN	mm	kpa		°C	kpa	kpa				mm	mm	%	mm	mm	mm	Schedule >>	mm	OK / NOT OK	OK / NOT OK	OK / NOT OK
	½	15	21.3	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	0.10	6.10	6.97	XXS	7.47	OK	OK	OK
	¾	20	26.7	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	0.12	6.12	7.00	XXS	7.82	OK	OK	OK
	1	25	33.4	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	0.15	6.15	7.03	XXS	9.09	OK	OK	OK
	1½	40	48.3	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	0.22	6.22	7.11	XXS	10.15	OK	OK	OK
	2	50	60.3	1,900	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	0.28	6.28	7.17	160	8.74	OK	OK	OK

WALL THICKNESS CALCULATION FOR MATERIAL LTC A333 Gr. 6 (B23-150#) PER ASME B 31.3-2018

	No.		DESIGN DATA SUMMARY	NOTATION		INPUT	UNIT	REMARK
	1	Design	Design Temperature		>>	38	°C	ATM Temp = 38°C
	2	Condition	Design Pressure	P	>>	19.6	Bar	
	3	Pipe Data	Pipe Material		>>	A333 Gr. 6		
	4		Pipe Allowable Stress at Design Temp. 38°C	S _{oper}	=	206.84	MPa	
			Pipe Allowable Stress at ATM Temp. 38°C	S _{atm}	=	137.90	MPa	
	5	Allowance	Corrosion Allowance, Bore	C.A	>>	3.0	mm	
	6		Thread Allowance, Bore	M.A	>>	0.00	mm	
	7	Factor	Longitudinal Weld Joints Factor	E	>>	1.00		
	8		Temperature Coefficient	Y	>>	0.4		
	9		Weld Joint Strength Reduction Factor	W	>>	1.00		
	10	Tolerance	Mill Tolerance		>>	12.5	%	

PROJECT TITLE : Front End Engineering Design for Gas Gathering Pipelines (Phase1)																						
PROJECT No : 11998A																						
PIPE CLASS : B23																						
	Pipe Size		Outside Diameter	Design Pressure	Material	Temp	Allowable Stress	Yield Stress	E Factor	W (Weld Joint strength Reduction Factor)	Y Coefficient	Corrosion Allowance	Mech. Allowance (Thread/Groove Depth)	Mill. Tol	Pressure Design Thickness	Required Thickness	Minimum Thickness	Selected Schedule / Thickness		Selected Wall Thickness Validation	Para 304.1.2 (a) Equation (3a) Validation	
			D	P			S _o	S _y	E	W	Y	C.A	M.A	Tol.	t	t _m	T	T _s		T _s ≥ T	t < D/6	P _{/SE} ≤ 0.385
	NPS	DN	mm	kpa		°C	kpa	kpa				mm	mm	%	mm	mm	mm	Schedule >>	mm	OK / NOT OK	OK / NOT OK	OK / NOT OK
	½	15	21.3	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	0.10	3.10	3.54	160	4.78	OK	OK	OK
	¾	20	26.7	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	0.13	3.13	3.57	160	5.56	OK	OK	OK
	1	25	33.4	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	0.16	3.16	3.61	160	6.35	OK	OK	OK
	1½	40	48.3	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	0.23	3.23	3.69	160	7.14	OK	OK	OK
	2	50	60.3	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	0.28	3.28	3.75	XS	5.54	OK	OK	OK
	3	80	88.9	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	0.42	3.42	3.91	STD	5.49	OK	OK	OK
	4	100	114.3	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	0.54	3.54	4.05	STD	6.02	OK	OK	OK
	6	150	168.3	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	0.79	3.79	4.34	STD	7.11	OK	OK	OK
	8	200	219.1	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	1.03	4.03	4.61	20	6.35	OK	OK	OK
	10	250	273.1	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	1.29	4.29	4.90	20	6.35	OK	OK	OK
	12	300	323.9	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	1.53	4.53	5.18	30	8.38	OK	OK	OK
	14	350	355.6	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	1.68	4.68	5.35	20	7.92	OK	OK	OK
	16	400	406.4	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	1.92	4.92	5.62	20	7.92	OK	OK	OK
	18	450	457.0	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	2.16	5.16	5.89	20	7.92	OK	OK	OK
	20	500	508.0	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	2.40	5.40	6.17	STD	9.53	OK	OK	OK
	24	600	610.0	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	2.88	5.88	6.72	STD	9.53	OK	OK	OK
	26	650	660.0	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	3.12	6.12	6.99	STD	9.53	OK	OK	OK
	28	700	711.0	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	3.36	6.36	7.26	STD	9.53	OK	OK	OK
	30	750	762.0	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	3.60	6.60	7.54	STD	9.53	OK	OK	OK
	32	800	813.0	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	3.84	6.84	7.81	40	10.31	OK	OK	OK
	34	850	864.0	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	4.08	7.08	8.09	XS	12.7	OK	OK	OK
	36	900	914.0	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	4.31	7.31	8.36	XS	12.7	OK	OK	OK
	38	950	965.0	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	4.55	7.55	8.63	XS	12.7	OK	OK	OK
	40	1000	1016.0	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	4.80	7.80	8.91	XS	12.7	OK	OK	OK
	42	1050	1067.0	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	5.04	8.04	9.18	XS	12.7	OK	OK	OK
	44	1100	1118.0	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	5.28	8.28	9.46	XS	12.7	OK	OK	OK
	46	1150	1168.0	1,960	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	3.0	0.00	12.5	5.51	8.51	9.73	XS	12.7	OK	OK	OK

WALL THICKNESS CALCULATION FOR MATERIAL API 5LGr B (C03 -900#) PER ASME B 31.3-2018

	No.	DESIGN DATA SUMMARY		NOTATION		INPUT	UNIT	REMARK
	1	Design	Design Temperature		>>	38	°C	ATM Temp = 38°C
	2	Condition	Design Pressure	P	>>	153.2	Bar	
	3	Pipe Data	Pipe Material		>>	API 5L Gr. B		
	4		Pipe Allowable Stress at Design Temp. 38°C	S _{oper}	=	206.84	MPa	
			Pipe Allowable Stress at ATM Temp. 38°C	S _{atm}	=	137.90	MPa	
	5	Allowance	Corrosion Allowance, Bore	C.A	>>	3.0	mm	
	6		Thread Allowance, Bore	M.A	>>	0.00	mm	
	7	Factor	Longitudinal Weld Joints Factor	E	>>	1.00		
	8		Temperature Coefficient	Y	>>	0.4		
	9		Weld Joint Strength Reduction Factor	W	>>	1.00		
	10	Tolerance	Mill Tolerance		>>	12.5	%	

PROJECT TITLE : Front End Engineering Design for Gas Gathering Pipelines (Phase1)

PROJECT No : 11998A

PIPE CLASS : C03 (1/2" TO 16")

[illegible]

WALL THICKNESS CALCULATION FOR MATERIAL API 5L Gr. X60 (C03 -900#) PER ASME B 31.3-2018

	No.		DESIGN DATA SUMMARY	NOTATION		INPUT	UNIT	REMARK
	1	Design	Design Temperature		>>	38	°C	ATM Temp = 38°C
	2	Condition	Design Pressure	P	>>	153.2	Bar	
	3	Pipe Data	Pipe Material		>>	API 5L Gr. X60		
	4		Pipe Allowable Stress at Design Temp. 38°C	S _{oper}	=	206.84	MPa	
			Pipe Allowable Stress at ATM Temp. 38°C	S _{atm}	=	172.37	MPa	
	5	Allowance	Corrosion Allowance, Bore	C.A	>>	3.0	mm	
	6		Thread Allowance, Bore	M.A	>>	0.00	mm	
	7	Factor	Longitudinal Weld Joints Factor	E	>>	1.00		
	8		Temperature Coefficient	Y	>>	0.4		
	9		Weld Joint Strength Reduction Factor	W	>>	1.00		
	10	Tolerance	Mill Tolerance		>>	12.5	%	

PROJECT TITLE : <u>Front End Engineering Design for Gas Gathering Pipelines (Phase1)</u>																						
PROJECT No : <u>11998A</u>																						
PIPE CLASS : <u>C03 (18" TO 46")</u>																						
	Pipe Size		Outside Diameter	Design Pressure	Material	Temp	Allowable Stress	Yield Stress	E Factor	W (Weld Joint Strength Reduction Factor)	Y Coefficient	Corrosion Allowance	Mech. Allowance (Thread/Groove Depth)	Mill. Tol	Pressure Design Thickness	Required Thickness	Minimum Thickness	Selected Schedule / Thickness		Selected Wall Thickness Validation	Para 304.1.2 (a) Equation (3a) Validation	
			D	P			S _o	S _y	E	W	Y	C.A	M.A	Tol.	t	t _m	T	T _s		T _s ≥ T	t < D/6	P _i /S _E ≤ 0.385
	NPS	DN	mm	kpa		°C	kpa	kpa				mm	mm	%	mm	mm	mm	Schedule >>	mm	OK / NOT OK	OK / NOT OK	OK / NOT OK
	18	450	457.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	3.0	0.00	12.5	16.44	19.44	22.21	100	29.36	OK	OK	OK
	20	500	508.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	3.0	0.00	12.5	18.27	21.27	24.31	100	32.54	OK	OK	OK
	24	600	610.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	3.0	0.00	12.5	21.94	24.94	28.50	100	38.89	OK	OK	OK
	26	650	660.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	3.0	0.00	12.5	23.74	26.74	30.56	-	34.1	OK	OK	OK
	28	700	711.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	3.0	0.00	12.5	25.57	28.57	32.65	-	36.5	OK	OK	OK
	30	750	762.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	3.0	0.00	12.5	27.41	30.41	34.75	-	38.9	OK	OK	OK
	32	800	813.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	3.0	0.00	12.5	29.24	32.24	36.85	-	41.2	OK	OK	OK
	34	850	864.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	3.0	0.00	12.5	31.08	34.08	38.94	-	43.6	OK	OK	OK
	36	900	914.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	3.0	0.00	12.5	32.87	35.87	41.00	-	46	OK	OK	OK
	38	950	965.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	3.0	0.00	12.5	34.71	37.71	43.10	-	48.3	OK	OK	OK
	40	1000	1016.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	3.0	0.00	12.5	36.54	39.54	45.19	-	50.7	OK	OK	OK
	42	1050	1067.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	3.0	0.00	12.5	38.38	41.38	47.29	-	53.1	OK	OK	OK
	44	1100	1118.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	3.0	0.00	12.5	40.21	43.21	49.38	-	55.4	OK	OK	OK
	46	1150	1168.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	3.0	0.00	12.5	42.01	45.01	51.44	-	57.8	OK	OK	OK

WALL THICKNESS CALCULATION FOR MATERIAL API 5LGr B (C06 -900#) PER ASME B 31.3-2018

	No.		DESIGN DATA SUMMARY	NOTATION		INPUT	UNIT	REMARK
	1	Design	Design Temperature		>>	38	°C	ATM Temp = 38°C
	2	Condition	Design Pressure	P	>>	153.2	Bar	
	3	Pipe Data	Pipe Material		>>	API 5L Gr. B		
	4		Pipe Allowable Stress at Design Temp. 38°C	S _{oper}	=	206.84	MPa	
			Pipe Allowable Stress at ATM Temp. 38°C	S _{atm}	=	137.90	MPa	
	5	Allowance	Corrosion Allowance, Bore	C.A	>>	6.0	mm	
	6		Thread Allowance, Bore	M.A	>>	0.00	mm	
	7	Factor	Longitudinal Weld Joints Factor	E	>>	1.00		
	8		Temperature Coefficient	Y	>>	0.4		
	9		Weld Joint Strength Reduction Factor	W	>>	1.00		
	10	Tolerance	Mill Tolerance		>>	12.5	%	

PROJECT TITLE : [Front End Engineering Design for Gas Gathering Pipelines \(Phase1\)](#)
PROJECT No : [11998A](#)
PIPE CLASS : [C06 \(1/2" TO 16"\)](#)

	Pipe Size		Outside Diameter	Design Pressure	Material	Temp	Allowable Stress	Yield Stress	E Factor	W (Weld Joint Strength Reduction Factor)	Y Coefficient	Corrosion Allowance	Mech. Allowance (Thread/Groove Depth)	Mill. Tol	Pressure Design Thickness	Required Thickness	Minimum Thickness	Selected Schedule / Thickness		Selected Wall Thickness Validation	Para 304.1.2 (a) Equation (3a) Validation		
			D	P			S _o	S _y	E	W	Y	C.A	M.A	ToL.	t	t _m	T	T _s		T _s ≥ T	t < D/6	^P / _{SE} ≤ 0.385	
	NPS	DN	mm	kpa		°C	kpa	kpa				mm	mm	%	mm	mm	mm	Schedule >>	mm	OK / NOT OK	OK / NOT OK	OK / NOT OK	
	½	15	21.3	15,320	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	0.77	6.77	7.73	XXS	7.47	NOT OK	OK	OK	
	¾	20	26.7	15,320	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	0.96	6.96	7.95	XXS	7.82	NOT OK	OK	OK	
	1	25	33.4	15,320	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	1.20	7.20	8.23	XXS	9.09	OK	OK	OK	
	1½	40	48.3	15,320	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	1.74	7.74	8.84	XXS	10.15	OK	OK	OK	
	2	50	60.3	15,320	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	2.17	8.17	9.34	XXS	11.07	OK	OK	OK	
	3	80	88.9	15,320	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	3.20	9.20	10.51	XXS	15.24	OK	OK	OK	
	4	100	114.3	15,320	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	4.11	10.11	11.56	XXS	17.12	OK	OK	OK	
	6	150	168.3	15,320	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	6.05	12.05	13.78	160	18.26	OK	OK	OK	
	8	200	219.1	15,320	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	7.88	13.88	15.86	140	20.62	OK	OK	OK	
	10	250	273.1	15,320	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	9.82	15.82	18.08	XXS	25.4	OK	OK	OK	
	12	300	323.9	15,320	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	11.65	17.65	20.17	140	28.58	OK	OK	OK	
	14	350	355.6	15,320	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	12.79	18.79	21.47	140	31.75	OK	OK	OK	
	16	400	406.4	15,320	API 5L Gr. B	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	14.62	20.62	23.56	140	36.53	OK	OK	OK	

WALL THICKNESS CALCULATION FOR MATERIAL API 5L Gr. X60 (C06 -900#) PER ASME B 31.3-2018

	No.		DESIGN DATA SUMMARY	NOTATION		INPUT	UNIT	REMARK
	1	Design	Design Temperature		>>	38	°C	ATM Temp = 38°C
	2	Condition	Design Pressure	P	>>	153.2	Bar	
	3	Pipe Data	Pipe Material		>>	API 5L Gr. X60		
	4		Pipe Allowable Stress at Design Temp. 38°C	S _{oper}	=	206.84	MPa	
			Pipe Allowable Stress at ATM Temp. 38°C	S _{atm}	=	172.37	MPa	
	5	Allowance	Corrosion Allowance, Bore	C.A	>>	6.0	mm	
	6		Thread Allowance, Bore	M.A	>>	0.00	mm	
	7	Factor	Longitudinal Weld Joints Factor	E	>>	1.00		
	8		Temperature Coefficient	Y	>>	0.4		
	9		Weld Joint Strength Reduction Factor	W	>>	1.00		
	10	Tolerance	Mill Tolerance		>>	12.5	%	

PROJECT TITLE : Front End Engineering Design for Gas Gathering Pipelines (Phase1)
PROJECT No : 11998A
PIPE CLASS : C06 (18" TO 46")

	Pipe Size		Outside Diameter	Design Pressure	Material	Temp	Allowable Stress	Yield Stress	E Factor	W (Weld Joint Strength Reduction Factor)	Y Coefficient	Corrosion Allowance	Mech. Allowance (Thread/Groove Depth)	Mill. Tol	Pressure Design Thickness	Required Thickness	Minimum Thickness	Selected Schedule / Thickness		Selected Wall Thickness Validation	Para 304.1.2 (a) Equation (3a) Validation	
			D	P			S _o	S _y	E	W	Y	C.A	M.A	Tol.	t	t _m	T	T _s		T _s ≥ T	t < D/6	P _{/SE} ≤ 0.385
	NPS	DN	mm	kpa		°C	kpa	kpa				mm	mm	%	mm	mm	mm	Schedule >>	mm	OK / NOT OK	OK / NOT OK	OK / NOT OK
	18	450	457.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	6.0	0.00	12.5	16.44	22.44	25.64	100	29.36	OK	OK	OK
	20	500	508.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	6.0	0.00	12.5	18.27	24.27	27.74	100	32.54	OK	OK	OK
	24	600	610.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	6.0	0.00	12.5	21.94	27.94	31.93	100	38.89	OK	OK	OK
	26	650	660.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	6.0	0.00	12.5	23.74	29.74	33.99	-	37.4	OK	OK	OK
	28	700	711.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	6.0	0.00	12.5	25.57	31.57	36.08	-	39.7	OK	OK	OK
	30	750	762.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	6.0	0.00	12.5	27.41	33.41	38.18	-	42.1	OK	OK	OK
	32	800	813.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	6.0	0.00	12.5	29.24	35.24	40.28	-	44.5	OK	OK	OK
	34	850	864.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	6.0	0.00	12.5	31.08	37.08	42.37	-	46.9	OK	OK	OK
	36	900	914.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	6.0	0.00	12.5	32.87	38.87	44.43	-	49.2	OK	OK	OK
	38	950	965.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	6.0	0.00	12.5	34.71	40.71	46.52	-	51.6	OK	OK	OK
	40	1000	1016.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	6.0	0.00	12.5	36.54	42.54	48.62	-	54	OK	OK	OK
	42	1050	1067.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	6.0	0.00	12.5	38.38	44.38	50.72	-	56.3	OK	OK	OK
	44	1100	1118.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	6.0	0.00	12.5	40.21	46.21	52.81	-	58.7	OK	OK	OK
	46	1150	1168.0	15,320	API 5L Gr. X60	38	206,844	413,688	1.00	1.00	0.4	6.0	0.00	12.5	42.01	48.01	54.87	-	61.1	OK	OK	OK

WALL THICKNESS CALCULATION FOR MATERIAL LTC A333 Gr. 6 (C23-900#) PER ASME B 31.3-2018

	No.		DESIGN DATA SUMMARY	NOTATION		INPUT	UNIT	REMARK
	1	Design	Design Temperature		>>	38	°C	ATM Temp = 38°C
	2	Condition	Design Pressure	P	>>	153.2	Bar	
	3	Pipe Data	Pipe Material		>>	A333 Gr. 6		
	4		Pipe Allowable Stress at Design Temp. 38°C	S _{oper}	=	206.84	MPa	
			Pipe Allowable Stress at ATM Temp. 38°C	S _{atm}	=	137.90	MPa	
	5	Allowance	Corrosion Allowance, Bore	C.A	>>	6.0	mm	
	6		Thread Allowance, Bore	M.A	>>	0.00	mm	
	7	Factor	Longitudinal Weld Joints Factor	E	>>	1.00		
	8		Temperature Coefficient	Y	>>	0.4		
	9		Weld Joint Strength Reduction Factor	W	>>	1.00		
	10	Tolerance	Mill Tolerance		>>	12.5	%	

PROJECT TITLE : Front End Engineering Design for Gas Gathering Pipelines (Phase1)																						
PROJECT No : 11998A																						
PIPE CLASS : C23																						
	Pipe Size		Outside Diameter	Design Pressure	Material	Temp	Allowable Stress	Yield Stress	E Factor	W (Weld Joint strength Reduction Factor)	Y Coefficient	Corrosion Allowance	Mech. Allowance (Thread/Groove Depth)	Mill. Tol	Pressure Design Thickness	Required Thickness	Minimum Thickness	Selected Schedule / Thickness		Selected Wall Thickness Validation	Para 304.1.2 (a) Equation (3a) Validation	
			D	P			S _o	S _y	E	W	Y	C.A	M.A	Tol.	t	t _m	T	T _s		T _s ≥ T	t < D/6	P _{/SE} ≤ 0.385
	NPS	DN	mm	kpa		°C	kpa	kpa				mm	mm	%	mm	mm	mm	Schedule >>	mm	OK / NOT OK	OK / NOT OK	OK / NOT OK
	½	15	21.3	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	0.77	6.77	7.73	160	4.78	NOT OK	OK	OK
	¾	20	26.7	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	0.96	6.96	7.95	160	5.56	NOT OK	OK	OK
	1	25	33.4	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	1.20	7.20	8.23	160	6.35	NOT OK	OK	OK
	1½	40	48.3	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	1.74	7.74	8.84	160	7.14	NOT OK	OK	OK
	2	50	60.3	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	2.17	8.17	9.34	160	8.74	NOT OK	OK	OK
	3	80	88.9	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	3.20	9.20	10.51	160	11.13	OK	OK	OK
	4	100	114.3	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	4.11	10.11	11.56	120	11.13	NOT OK	OK	OK
	6	150	168.3	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	6.05	12.05	13.78	120	14.27	OK	OK	OK
	8	200	219.1	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	7.88	13.88	15.86	120	18.26	OK	OK	OK
	10	250	273.1	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	9.82	15.82	18.08	120	21.44	OK	OK	OK
	12	300	323.9	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	11.65	17.65	20.17	XXS	25.4	OK	OK	OK
	14	350	355.6	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	12.79	18.79	21.47	120	27.79	OK	OK	OK
	16	400	406.4	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	14.62	20.62	23.56	120	30.96	OK	OK	OK
	18	450	457.0	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	16.44	22.44	25.64	120	34.93	OK	OK	OK
	20	500	508.0	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	18.27	24.27	27.74	120	38.1	OK	OK	OK
	24	600	610.0	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	21.94	27.94	31.93	100	38.89	OK	OK	OK
	26	650	660.0	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	23.74	29.74	33.99	-	41.5	OK	OK	OK
	28	700	711.0	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	25.57	31.57	36.08	-	44.4	OK	OK	OK
	30	750	762.0	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	27.41	33.41	38.18	-	47.4	OK	OK	OK
	32	800	813.0	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	29.24	35.24	40.28	-	50.3	OK	OK	OK
	34	850	864.0	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	31.08	37.08	42.37	-	53.2	OK	OK	OK
	36	900	914.0	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	32.87	38.87	44.43	-	56.2	OK	OK	OK
	38	950	965.0	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	34.71	40.71	46.52	-	59.1	OK	OK	OK
	40	1000	1016.0	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	36.54	42.54	48.62	-	62	OK	OK	OK
	42	1050	1067.0	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	38.38	44.38	50.72	-	65	OK	OK	OK
	44	1100	1118.0	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	40.21	46.21	52.81	-	67.9	OK	OK	OK
	46	1150	1168.0	15,320	A333 Gr. 6	38	206,844	241,318	1.00	1.00	0.4	6.0	0.00	12.5	42.01	48.01	54.87	-	70.9	OK	OK	OK

WALL THICKNESS CALCULATION FOR MATERIAL SS A312 TP316L (C50-900#) PER ASME B 31.3-2018

	No.		DESIGN DATA SUMMARY	NOTATION		INPUT	UNIT	REMARK
	1	Design	Design Temperature		>>	38	°C	ATM Temp = 38°C
	2	Condition	Design Pressure	P	>>	124.1	Bar	
	3	Pipe Data	Pipe Material		>>	A312 TP316L		
	4		Pipe Allowable Stress at Design Temp. 38°C	S _{oper}	=	206.84	MPa	
			Pipe Allowable Stress at ATM Temp. 38°C	S _{atm}	=	115.14	MPa	
	5	Allowance	Corrosion Allowance, Bore	C.A	>>	0.0	mm	
	6		Thread Allowance, Bore	M.A	>>	0.00	mm	
	7	Factor	Longitudinal Weld Joints Factor	E	>>	1.00		
	8		Temperature Coefficient	Y	>>	0.4		
	9		Weld Joint Strength Reduction Factor	W	>>	1.00		
	10	Tolerance	Mill Tolerance		>>	12.5	%	

PROJECT TITLE : Front End Engineering Design for Gas Gathering Pipelines (Phase1)
PROJECT No : 11998A
PIPE CLASS : C50

	Pipe Size		Outside Diameter	Design Pressure	Material	Temp	Allowable Stress	Yield Stress	E Factor	W (Weld Joint Strength Reduction Factor)	Y Coefficient	Corrosion Allowance	Mech. Allowance (Thread/Groove Depth)	Mill. Tol	Pressure Design Thickness	Required Thickness	Minimum Thickness	Selected Wall Thickness Validation	Para 304.1.2 (a) Equation (3a) Validation	
			D	P			S _o	S _y	E	W	Y	C.A	M.A	Tol.	t	t _m	T	T _s ≥ T	t < D/6	P _s /SE ≤ 0.385
	NPS	DN	mm	kpa		°C	kpa	kpa				mm	mm	%	mm	mm	mm	OK / NOT OK	OK / NOT OK	OK / NOT OK
	½	15	21.3	12,410	A312 TP316L	38	206,844	172,370	1.00	1.00	0.4	0.0	0.00	12.5	0.62	0.62	0.71	OK	OK	OK
	¾	20	26.7	12,410	A312 TP316L	38	206,844	172,370	1.00	1.00	0.4	0.0	0.00	12.5	0.78	0.78	0.89	OK	OK	OK
	1	25	33.4	12,410	A312 TP316L	38	206,844	172,370	1.00	1.00	0.4	0.0	0.00	12.5	0.98	0.98	1.12	OK	OK	OK
	1½	40	48.3	12,410	A312 TP316L	38	206,844	172,370	1.00	1.00	0.4	0.0	0.00	12.5	1.41	1.41	1.62	OK	OK	OK
	2	50	60.3	12,410	A312 TP316L	38	206,844	172,370	1.00	1.00	0.4	0.0	0.00	12.5	1.77	1.77	2.02	OK	OK	OK

WALL THICKNESS CALCULATION FOR MATERIAL DUPLEX S31803 (C70-900#) PER ASME B 31.3-2018

	No.		DESIGN DATA SUMMARY	NOTATION		INPUT	UNIT	REMARK
	1	Design	Design Temperature		>>	38	°C	ATM Temp = 38°C
	2	Condition	Design Pressure	P	>>	155.1	Bar	
	3	Pipe Data	Pipe Material		>>	A790 S31803		
	4		Pipe Allowable Stress at Design Temp. 38°C	S _{oper}	=	206.84	MPa	
			Pipe Allowable Stress at ATM Temp. 38°C	S _{atm}	=	206.84	MPa	
	5	Allowance	Corrosion Allowance, Bore	C.A	>>	0.0	mm	
	6		Thread Allowance, Bore	M.A	>>	0.00	mm	
	7	Factor	Longitudinal Weld Joints Factor	E	>>	1.00		
	8		Temperature Coefficient	Y	>>	0.4		
	9		Weld Joint Strength Reduction Factor	W	>>	1.00		
	10	Tolerance	Mill Tolerance		>>	8	%	

PROJECT TITLE : Front End Engineering Design for Gas Gathering Pipelines (Phase1)																						
PROJECT No : 11998A																						
PIPE CLASS : C70 DUPLEX																						
	Pipe Size		Outside Diameter	Design Pressure	Material	Temp	Allowable Stress	Yield Stress	E Factor	W (Weld Joint Strength Reduction Factor)	Y Coefficient	Corrosion Allowance	Mech. Allowance (Thread/Groove Depth)	Mill. Tol	Pressure Design Thickness	Required Thickness	Minimum Thickness	Selected Schedule / Thickness		Selected Wall Thickness Validation	Para 304.1.2 (a) Equation (3a) Validation	
			D	P			S _o	S _y	E	W	Y	C.A	M.A	Tol.	t	t _m	T	T _s		T _s ≥ T	t < D/6	P / S _E ≤ 0.385
	NPS	DN	mm	kpa		°C	kpa	kpa				mm	mm	%	mm	mm	mm	Schedule >>	mm	OK / NOT OK	OK / NOT OK	OK / NOT OK
	½	15	21.3	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	0.78	0.78	0.84	80S	3.73	OK	OK	OK
	¾	20	26.7	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	0.97	0.97	1.06	80S	3.91	OK	OK	OK
	1	25	33.4	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	1.22	1.22	1.32	80S	4.55	OK	OK	OK
	1½	40	48.3	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	1.76	1.76	1.91	80S	5.08	OK	OK	OK
	2	50	60.3	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	2.19	2.19	2.39	80S	5.54	OK	OK	OK
	3	80	88.9	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	3.24	3.24	3.52	40S	5.49	OK	OK	OK
	4	100	114.3	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	4.16	4.16	4.52	40S	6.02	OK	OK	OK
	6	150	168.3	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	6.13	6.13	6.66	40S	7.11	OK	OK	OK
	8	200	219.1	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	7.98	7.98	8.67	80S	12.7	OK	OK	OK
	10	250	273.1	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	9.94	9.94	10.81	80S	12.7	OK	OK	OK
	12	300	323.9	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	11.79	11.79	12.82	-	13.5	OK	OK	OK
	14	350	355.6	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	12.94	12.94	14.07	-	14.8	OK	OK	OK
	16	400	406.4	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	14.79	14.79	16.08	-	16.9	OK	OK	OK
	18	450	457.0	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	16.63	16.63	18.08	-	18.1	OK	OK	OK
	20	500	508.0	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	18.49	18.49	20.10	-	20.1	OK	OK	OK
	24	600	610.0	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	22.20	22.20	24.13	-	24.2	OK	OK	OK
	26	650	660.0	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	24.02	24.02	26.11	-	26.2	OK	OK	OK
	28	700	711.0	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	25.88	25.88	28.13	-	28.2	OK	OK	OK
	30	750	762.0	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	27.74	27.74	30.15	-	30.2	OK	OK	OK
	32	800	813.0	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	29.59	29.59	32.17	-	32.2	OK	OK	OK
	34	850	864.0	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	31.45	31.45	34.18	-	34.2	OK	OK	OK
	36	900	914.0	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	33.27	33.27	36.16	-	36.2	OK	OK	OK
	38	950	965.0	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	35.13	35.13	38.18	-	38.2	OK	OK	OK
	40	1000	1016.0	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	36.98	36.98	40.20	-	40.3	OK	OK	OK
	42	1050	1067.0	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	38.84	38.84	42.22	-	42.3	OK	OK	OK
	44	1100	1118.0	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	40.70	40.70	44.23	-	44.3	OK	OK	OK
	46	1150	1168.0	15,510	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	8.0	42.52	42.52	46.21	-	46.3	OK	OK	OK

WALL THICKNESS CALCULATION FOR MATERIAL DUPLEX S31803 (D70-1500#) PER ASME B 31.3-2018

	No.		DESIGN DATA SUMMARY	NOTATION		INPUT	UNIT	REMARK
	1	Design	Design Temperature		>>	38	°C	ATM Temp = 38°C
	2	Condition	Design Pressure	P	>>	258.6	Bar	
	3	Pipe Data	Pipe Material		>>	A790 S31803		
	4		Pipe Allowable Stress at Design Temp. 38°C	S _{oper}	=	206.84	MPa	
			Pipe Allowable Stress at ATM Temp. 38°C	S _{atm}	=	206.84	MPa	
	5	Allowance	Corrosion Allowance, Bore	C.A	>>	0.0	mm	
	6		Thread Allowance, Bore	M.A	>>	0.00	mm	
	7	Factor	Longitudinal Weld Joints Factor	E	>>	1.00		
	8		Temperature Coefficient	Y	>>	0.4		
	9		Weld Joint Strength Reduction Factor	W	>>	1.00		
	10	Tolerance	Mill Tolerance		>>	7	%	

PROJECT TITLE : Front End Engineering Design for Gas Gathering Pipelines (Phase1)																						
PROJECT No : 11998A																						
PIPE CLASS : D70 DUPLEX																						
	Pipe Size		Outside Diameter	Design Pressure	Material	Temp	Allowable Stress	Yield Stress	E Factor	W (Weld Joint Strength Reduction Factor)	Y Coefficient	Corrosion Allowance	Mech. Allowance (Thread/Groove Depth)	Mill. Tol	Pressure Design Thickness	Required Thickness	Minimum Thickness	Selected Schedule / Thickness		Selected Wall Thickness Validation	Para 304.1.2 (a) Equation (3a) Validation	
			D	P			S _o	S _y	E	W	Y	C.A	M.A	Tol.	t	t _m	T	T _s		T _s ≥ T	t < D/6	P _{/SE} ≤ 0.385
	NPS	DN	mm	kpa		°C	kpa	kpa				mm	mm	%	mm	mm	mm	Schedule >>	mm	OK / NOT OK	OK / NOT OK	OK / NOT OK
	½	15	21.3	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	1.27	1.27	1.36	80S	3.73	OK	OK	OK
	¾	20	26.7	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	1.59	1.59	1.71	80S	3.91	OK	OK	OK
	1	25	33.4	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	1.99	1.99	2.14	80S	4.55	OK	OK	OK
	1½	40	48.3	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	2.88	2.88	3.09	80S	5.08	OK	OK	OK
	2	50	60.3	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	3.59	3.59	3.86	80S	5.54	OK	OK	OK
	3	80	88.9	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	5.29	5.29	5.69	80S	7.62	OK	OK	OK
	4	100	114.3	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	6.80	6.80	7.32	80S	8.56	OK	OK	OK
	6	150	168.3	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	10.02	10.02	10.77	-	11.5	OK	OK	OK
	8	200	219.1	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	13.04	13.04	14.03	-	14.9	OK	OK	OK
	10	250	273.1	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	16.26	16.26	17.48	-	18.6	OK	OK	OK
	12	300	323.9	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	19.28	19.28	20.73	-	22.1	OK	OK	OK
	14	350	355.6	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	21.17	21.17	22.76	-	24.2	OK	OK	OK
	16	400	406.4	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	24.19	24.19	26.02	-	27.7	OK	OK	OK
	18	450	457.0	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	27.21	27.21	29.25	-	29.6	OK	OK	OK
	20	500	508.0	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	30.24	30.24	32.52	-	32.9	OK	OK	OK
	24	600	610.0	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	36.32	36.32	39.05	-	39.5	OK	OK	OK
	26	650	660.0	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	39.29	39.29	42.25	-	42.8	OK	OK	OK
	28	700	711.0	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	42.33	42.33	45.51	-	46.1	OK	OK	OK
	30	750	762.0	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	45.36	45.36	48.78	-	49.4	OK	OK	OK
	32	800	813.0	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	48.40	48.40	52.04	-	52.6	OK	OK	OK
	34	850	864.0	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	51.44	51.44	55.31	-	55.9	OK	OK	OK
	36	900	914.0	25,860	A790 S31803	38	206,844	448,162	1.00	1.00	0.4	0.0	0.00	7.0	54.41	54.41	58.51	-	59.2	OK	OK	OK